

Assignment #01

Name: Fateh Ali

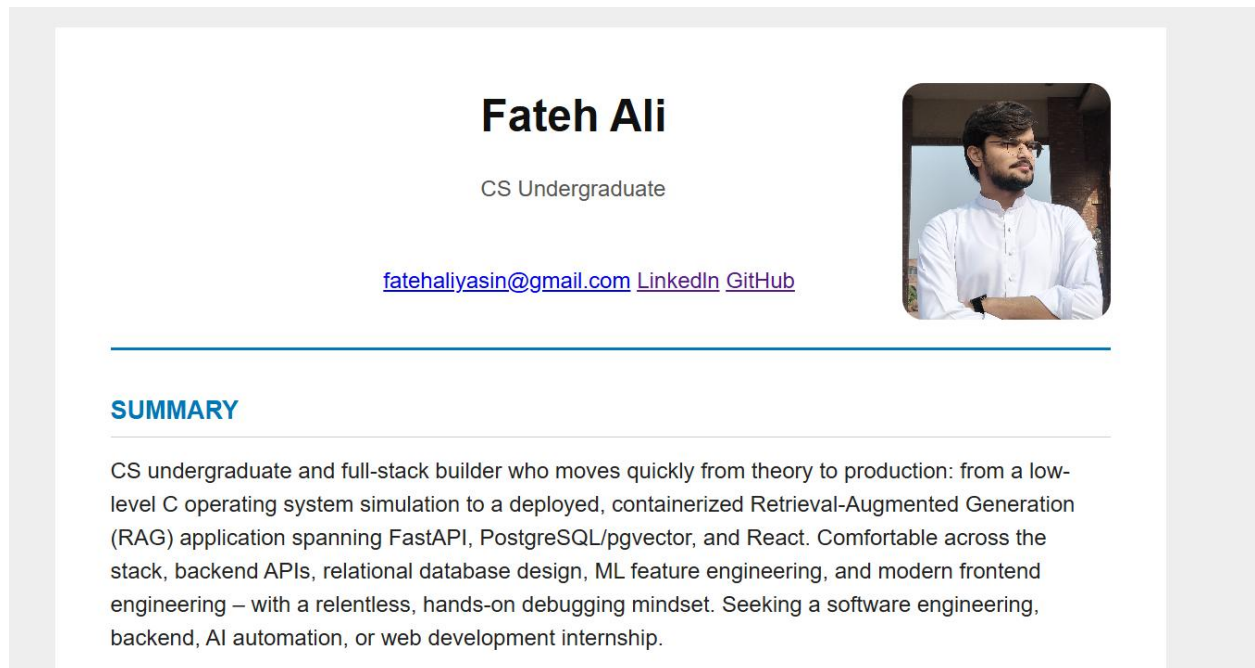
Reg. No: FA24-BCS-032

Assignment 1 — Your Personal CV Page

Difficulty: Foundational · **Covers:** Lectures 3–6 (HTML & HTML5, Semantic HTML and Forms, CSS Fundamentals, the CSS Box Model) · **Folder:** assignment-1/

Task: Build a single-page personal CV/résumé as a static HTML + CSS website — no JavaScript, no framework, just semantic HTML5 and hand-written CSS. This is where you prove you actually know the raw building blocks before anything is allowed to hide them from you.

Screenshots:



EDUCATION

BS Computer Science

COMSATS University Islamabad (Lahore Campus)

09/2024 – Present

TECHNICAL SKILLS

Languages & Concepts: Python, JavaScript, C++, SQL, Java, Object-Oriented Programming (OOP), Relational Database Design, RESTful APIs, ETL Pipelines, Web Scraping, Feature Engineering

Frameworks & Tools: FastAPI, React, Vite, Tailwind CSS, Docker, PostgreSQL, pgvector, SQLAlchemy 2.0, Alembic, JWT Authentication, Git/GitHub, Typer, CustomTkinter, BeautifulSoup4, Loguru, SSMS, Microsoft Office

CERTIFICATES

- **Python for Everybody Specialization** - University of Michigan (Coursera)
- **Python Crash Course** - Google (Coursera)
- **Technical Volunteer, ACM Student Chapter** - Organized and facilitated technical events and academic workshops

PROJECTS

RepoMind– Full-Stack RAG Application

Python, FastAPI, React

- Engineered and deployed a production-ready Retrieval-Augmented Generation (RAG) application from scratch, containerizing a FastAPI backend with a PostgreSQL + pgvector database using Docker, and managing schema evolution via SQLAlchemy 2.0 ORM and Alembic migrations.
- Built a custom document ingestion pipeline that chunks text with overlapping boundaries, and integrated the Google Gemini Embeddings API to perform cosine-similarity search and generate hallucination-free answers via Groq's Llama-3.1-8b-instant model.
- Built a responsive React + Vite + Tailwind CSS frontend with JWT-based session persistence and a Markdown-rendered chat interface; deployed live on Vercel (frontend) and Railway (backend).

Tech Pulse Data Pipeline

Python, BeautifulSoup, Typer

- Engineered an automated pipeline utilizing requests and BeautifulSoup4 to extract, filter, and score unstructured web data based on weighted keyword density.
- Architected a modular CLI with Typer and Loguru to orchestrate data transformation, secure CSV storage, and automated HTML email dispatch via SMTP.

AutoFile Pro

Python, CustomTkinter

- Built a desktop file-management application with a modern dark-mode GUI, featuring intelligent file organization by extension, size, or modification age.
- Implemented threaded backup execution with optional zip compression to keep the UI responsive during large file operations, and a configurable cleanup tool with safe delete-to-recycle-bin/archive handling.

Code:

Index.html:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0"
  />
    <title>Resume</title>
    <link rel="stylesheet" href="style.css" />
  </head>

  <body>
    <div id="main" class="resume">
      <header class="header">
        <div class="header-content">
          <h1>Fateh Ali</h1>
          <p>CS Undergraduate</p>
          <div class="urls">
            <ul id="creds">
              <li>
                <a href="mailto:fatehaliyasin@gmail.com" class="creds-
links"
                >fatehaliyasin@gmail.com</a>
              >
            </li>
            <li>
              <a
                href="https://linkedin.com/in/fateh-ali-072348352"
```

```

        class="creds-links"
        target="_blank"
        >LinkedIn</a>
    >
</li>
<li>
    <a
        href="https://github.com/FatehAli02"
        class="creds-links"
        target="_blank"
        >GitHub</a>
    >
</li>
</ul>
</div>
<div class="img">
    
</div>
</div>
</header>

```

<section>

<h2>Summary</h2>

<p>

CS undergraduate and full-stack builder who moves quickly from theory to production: from a low-level C operating system simulation to a deployed, containerized Retrieval-Augmented Generation (RAG) application spanning FastAPI, PostgreSQL/pgvector, and React. Comfortable across the stack, backend APIs, relational database design, ML feature engineering, and modern frontend engineering - with a relentless, hands-on debugging mindset. Seeking a software engineering, backend, AI automation, or web development internship.

</p>

</section>

<section>

<h2>Education</h2>

<h4>BS Computer Science</h4>

<p>COMSATS University Islamabad (Lahore Campus)</p>

```

    <p>09/2024 - Present</p>
  </section>

  <section>
    <h2>Technical Skills</h2>
    <p>
      <strong>Languages & Concepts: </strong>Python, JavaScript, C++,
SQL,
      Java, Object-Oriented Programming (OOP), Relational Database
Design,
      RESTful APIs, ETL Pipelines, Web Scraping, Feature Engineering
    </p>
    <p>
      <strong>Frameworks & Tools: </strong>FastAPI, React, Vite,
Tailwind
      CSS, Docker, PostgreSQL, pgvector, SQLAlchemy 2.0, Alembic, JWT
Authenticat ion, Git/GitHub, Typer, CustomTkinter,
BeautifulSoup4,
      Loguru, SSMS, Microsoft Office
    </p>
  </section>

  <section>
    <h2>Certificates</h2>
    <ul>
      <li>
        <b>Python for Everybody Specialization</b> - University of
Michigan
        (Coursera)
      </li>
      <li><b>Python Crash Course</b> - Google (Coursera)</li>
      <li>
        <b>Technical Volunteer, ACM Student Chapter</b> - Organized
and
        facilitated technical events and academic workshops
      </li>
    </ul>
  </section>

  <section>
    <h2>Projects</h2>
    <article class="project">
      <h3>RepoMind- Full-Stack RAG Application</h3>

```

```
<p class="technologies"><i>Python, FastAPI, React</i></p>
<ul>
  <li>
    Engineered and deployed a production-ready Retrieval-
Augmented
    Generation (RAG) application from scratch, containerizing a
    FastAPI backend with a PostgreSQL + pgvector database using
    Docker, and managing schema evolution via SQLAlchemy 2.0 ORM
and
    Alembic migrations.
  </li>
  <li>
    Built a custom document ingestion pipeline that chunks text
with
    overlapping boundaries, and integrated the Google Gemini
    Embeddings API to perform cosine-similarity search and
generate
    hallucination-free answers via Groq's llama-3.1-8b-instant
model.
  </li>
  <li>
    Built a responsive React + Vite + Tailwind CSS frontend with
    JWT-based session persistence and a Markdown-rendered chat
    interface; deployed live on Vercel (frontend) and Railway
    (backend).
  </li>
</ul>
</article>

<article class="project">
  <h3>Tech Pulse Data Pipeline</h3>
  <p class="technologies"><i>Python, BeautifulSoup, Typer</i></p>
  <ul>
    <li>
      Engineered an automated pipeline utilizing requests and
      BeautifulSoup4 to extract, filter, and score unstructured
web data
      based on weighted keyword density.
    </li>
    <li>
      Architected a modular CLI with Typer and Loguru to
orchestrate
```

```

        data transformation, secure CSV storage, and automated HTML
email
        dispatch via SMTP.
    </li>
</ul>
</article>

<article class="project">
    <h3>AutoFile Pro</h3>
    <p class="technologies"><i>Python, CustomTkinter</i></p>
    <ul>
        <li>
            Built a desktop file-management application with a modern
            dark-mode GUI, featuring intelligent file organization by
            extension, size, or modification age.
        </li>
        <li>
            Implemented threaded backup execution with optional zip
            compression to keep the UI responsive during large file
            operations, and a configurable cleanup tool with safe
            delete-to-recycle-bin/archive handling.
        </li>
    </ul>
</article>
</section>
</div>
</body>
</html>

```

Style.css:

```

* {
    box-sizing: border-box;
}

body {
    font-family: Arial, Helvetica, sans-serif;
    background-color: #eeeeee;
    color: #222222;
    line-height: 1.5;
    padding: 40px 20px;
    margin: 0;
}

```

```
}

.resume {
  width: 800px;
  margin: 30px auto;
  background-color: white;
  padding: 40px;
}

.header {
  border-bottom: 2px solid #0077b6;
  padding-bottom: 20px;
  margin-bottom: 30px;
}

.header-content {
  display: grid;
  grid-template-columns: 1fr 1.9fr auto;
  grid-template-rows: auto;
  row-gap: 5px;
}

.header-content h1 {
  grid-row: 1;
  grid-column: 2;
  margin: 0;
  font-size: 32px;
  color: #111111;
  padding-left: 70px;
}

.header-content p {
  grid-row: 2;
  grid-column: 2;
  margin: 5px 0 10px;
  color: #555555;
  size: 17px;
  padding-left: 70px;
}

.header-content .urls {
  grid-row: 3;
  grid-column: 2;
```

```
    align-self: end;
}

#creds {
  list-style: none;
  gap: 5px;
  padding: 0;
}

.header-content #creds li {
  display: inline;
}

.header-content .img {
  grid-column: 3;
  grid-row: 1 / span 3;
  display: flex;
  align-items: center;
  justify-content: flex-end;
}

.Profile_Pic {
  width: 150px;
  height: 170px;
  object-fit: cover;
  border-radius: 15px;
}

.credslinks {
  color: #0077b6;
  text-decoration: none;
  font-size: 14px;
}

.credslinks:hover {
  text-decoration: underline;
}

section {
  margin-bottom: 25px;
}

section h2 {
  margin-bottom: 12px;
  padding-bottom: 5px;
}
```

```
border-bottom: 1px solid #cccccc;
color: #0077b6;
font-size: 18px;
text-transform: uppercase;
}

section p {
  margin: 6px 0;
}

section h4 {
  margin: 8px 0 3px;
  font-size: 16px;
}

section ul {
  padding-left: 20px;
}

section li {
  margin-bottom: 6px;
}

.project {
  margin-bottom: 20px;
}

.project h3 {
  margin: 0 0 3px;
  font-size: 17px;
  color: #222222;
}

.technologies {
  margin: 0 0 8px;
  color: #666666;
  font-size: 14px;
}

.project ul {
  margin-top: 5px;
}
```

```
.project li {  
  line-height: 1.5;  
}
```